

Federated Learning Approach for Obstructive Sleep Apnea Prediction

Bachelor thesis presentation

Matej Vesel

UC Leuven Limburg & Result d.o.o

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Outline

Introduction

Project Background

Research and Development

Demo

Reflection

Project Approach

Questions and Answers

Introduction

- ▶ Team
 - ▶ **Manja Gorenc Novak**, *Head of AI and Senior Data Scientist*
 - ▶ Rok Vinder, *Data Analyst and Machine Learning Engineer*
- ▶ Result d.o.o
 - ▶ Digital solutions
- ▶ Context
 - ▶ eEarly
 - ▶ Health data
 - ▶ Obstructive sleep apnea

Project background

Problem statement and proposed solution

- ▶ Model consists of weights, imagine “knowledge”
- ▶ Problem statement
 - ▶ Standard machine learning
 - ▶ Data to the model (centralized)
 - ▶ Private (health) data
 - ▶ Regulations
 - ▶ Bandwidth
- ▶ Proposed solution
 - ▶ Proof-of-concept
 - ▶ Federated machine learning
 - ▶ Private data is stationary
 - ▶ Model to the data (decentralized)

Project background

Stakeholders and their needs

- ▶ **Developers**

- ▶ Understand code
- ▶ Interpret results
- ▶ Run the project

- ▶ *Patients*

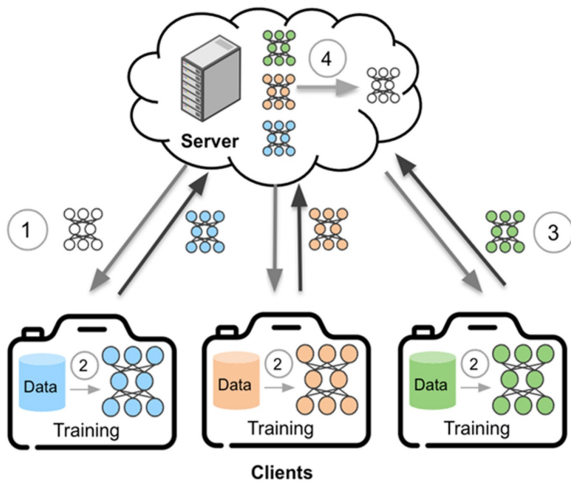
- ▶ **Data privacy**

- ▶ *Doctors*

- ▶ *Improved diagnosis*
- ▶ *Better tools*

Research and Development

Federated learning algorithm



Research and Development

Comparative study and technology decision

Flower

- ▶ beginner-friendly
- ▶ scalable

NVIDIA FLARE

- ▶ enterprise ready
- ▶ heavy for small projects

TensorFlow Federated

- ▶ complex
- ▶ the least integration

Demo

Reflection

Achieved objectives

- ▶ Larger dataset
- ▶ Approach comparison
- ▶ Metric difference ~ 0.03 (train)

Importance	Requirement	Done
Must have	Single-machine and non-containerized simulation of ap-nea predictions using federated learning.	Yes
Should have	Single-machine and containerized (with Docker) – –	Yes
Could have	Multi-machine and containerized (with Docker) – –	Yes
Could have	Multi-machine and containerized (on Kubernetes) – –	No
Could have	Differential privacy mechanism implemented.	No

Table: Red color represents MVP, orange and blue represent broader or ultimate scope. Bold requirements was shown in the demo.

Reflection

Improvements, and choice reflection

Improvements and weaknesses

- ▶ Evaluation metrics
- ▶ Separate networks
- ▶ Real TLS certification

Choice reflection

- ▶ Excellent
- ▶ No change

Reflection

Added value and impact

- ▶ Privacy (patients)
- ▶ Regulations (project managers, lawyers, developer)
- ▶ Better diagnosis tools (doctors)
- ▶ Less network usage (climate)
- ▶ Project opportunities (Result)
- ▶ Research community (scientists)

Project Approach

Development method, work division, and collaboration

Scrumban

- ▶ Agile mindset
- ▶ Daily meetings, backlog
- ▶ Typical kanban board

Work division

- ▶ MVP
 - ▶ Single-machine, non-containerized (plain Python)
- ▶ Additional (integration) features
 - ▶ Containerizing with Docker
 - ▶ Connecting multiple machines

Collaboaration

- ▶ Daily meetings, Slack, and Outlook

Thank you for your attention!

Questions and Answers